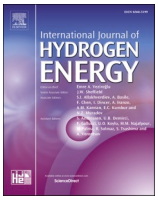
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Agrivoltaics and green hydrogen – Symbiosis of solar energy technologies for sustainable development of humanity☆

V.A. Panchenko [a,*], A.A. Kovalev [b], S. Chakraborty [c]

[a] Russian University of Transport, Obraztsova st., 9, Moscow, 127994, Russia
[b] Federal Scientific Agroengineering Center VIM, 1st Institutskiy proezd, 5, Moscow, 109428, Russia
[c] TIFAC- CORE, Vellore Institute of Technology (VIT), Vellore, India

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ABSTRACT

The article presents an overview of studies of agrivoltaic systems based on various types of photovoltaic modules, which have significant potential for the production and use of green hydrogen. The prerequisites for the active and effective implementation of agrivoltaics and green hydrogen in the modern world are considered, successful examples of the use of photovoltaic converters in agriculture and the production of green hydrogen are described, a classification of agrivoltaic systems is presented. The main advantages and disadvantages of using photovoltaic modules in agriculture are shown using examples of open, greenhouse and closed agrivoltaic systems. Modern agricultural machines and autonomous robots that are used in agrivoltaic systems are presented, and the feasibility of using green hydrogen in such devices is substantiated.

1. Introduction

The main global trend in the world energy sector is the gradual transition from fossil fuels in the energy and industry of the world's leading countries to renewable energy sources (RES) («global energy transition») [1–3], the priority goal of which is to reduce greenhouse gas (GHG) emissions in the production of electrical energy (EE) and thermal energy (TE). Hydrogen technologies play an important role in this process [4–6], since the energy sector accounts for more than two-thirds of GHG emissions [7,8]. «Green hydrogen» (GH) [9–13] in turn is the main driver of hydrogen energy today, which is produced using EE from RES converters [14,15] and solar plants (SP) [16,17], in particular.

The production of biohydrogen from organic waste (OW) using anaerobic digestion (AD) is also a promising technology [18,19] in the global energy transition. Such energy sources allow obtaining fuel and energy without harming the environment, and the used substrate in the form of OW is converted into high-quality fertilizers. The production of biohydrogen and biomethane from OW using AD can already today ensure the fulfillment of several important goals: the utilization of OW to obtain high-quality fertilizers, cost savings in the burial and discharge of industrial waste, the production of energy-intensive and valuable gases in the chemical and energy industries, and the sale of energy when using an electric generator on gaseous fuel [20–23].

Solar energy (SE) conversion is currently the fastest growing area of energy in the field of RES, while photovoltaic SE conversion has been used for quite some time all over the world due to the increase in the efficiency of photovoltaic converters (PVC) (Fig. 1a) [24] and the steadily decreasing cost of the specific power of photovoltaic modules (PVM) (Fig. 1b) [25].

At the same time, among all RES, SE is the most compatible with agricultural (ag/cul) activities [25]. The reasons for the reduction in the cost of polycrystalline PVMs are mainly associated with the improvement of materials, increased efficiency of photoelectric conversion, increased service life of PVCs and improved public policy [26]. Along with the deterioration of the environment, associated with a decrease in forest areas (Fig. 2a) [27], a noticeable increase in the world population is recorded (Fig. 2b) [28], which also leads to an increase in the area of agricultural land for the cultivation of ag/cul products.

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* Corresponding author.
E-mail address: pancheska@mail.ru (V.A. Panchenko).

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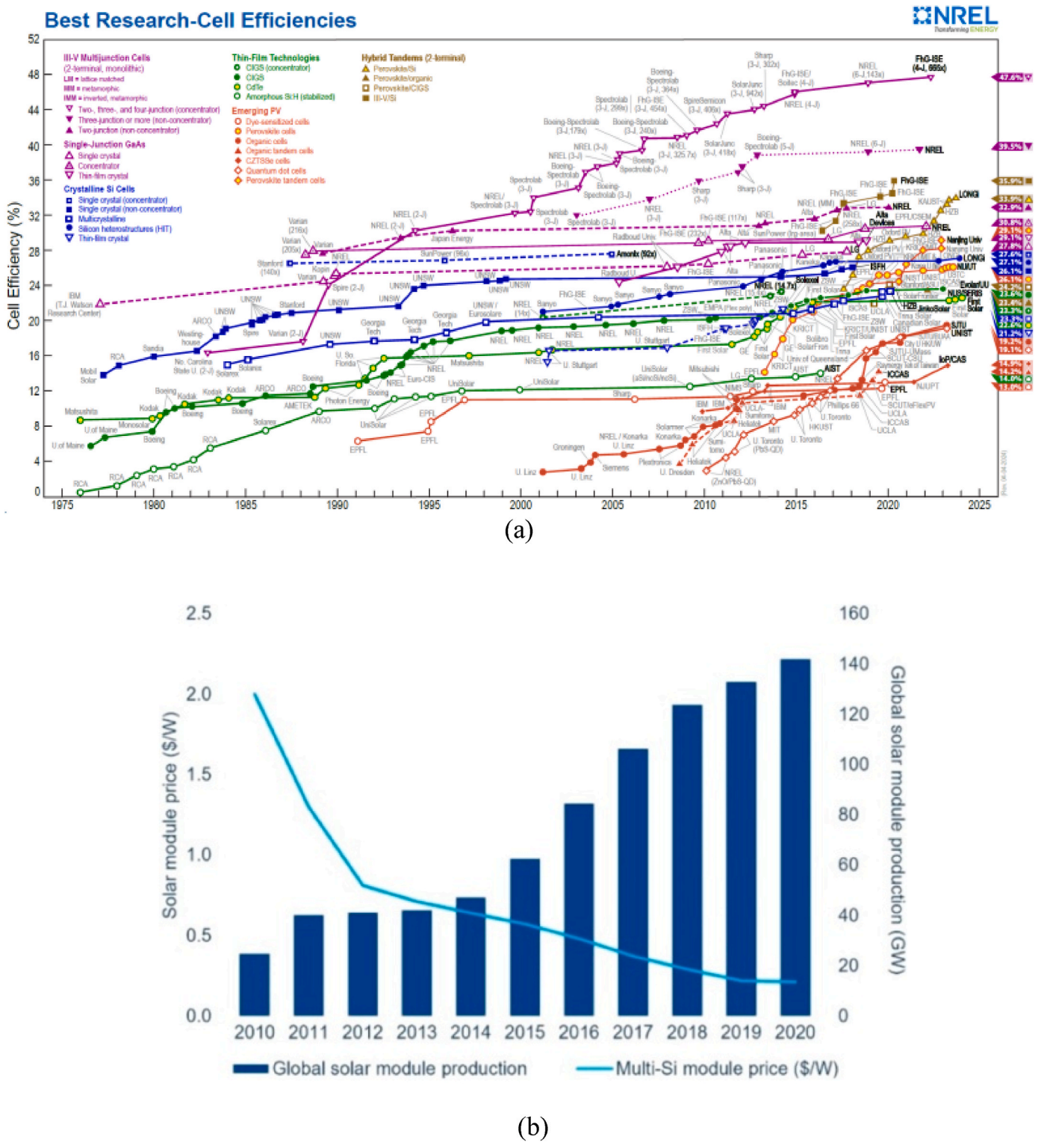
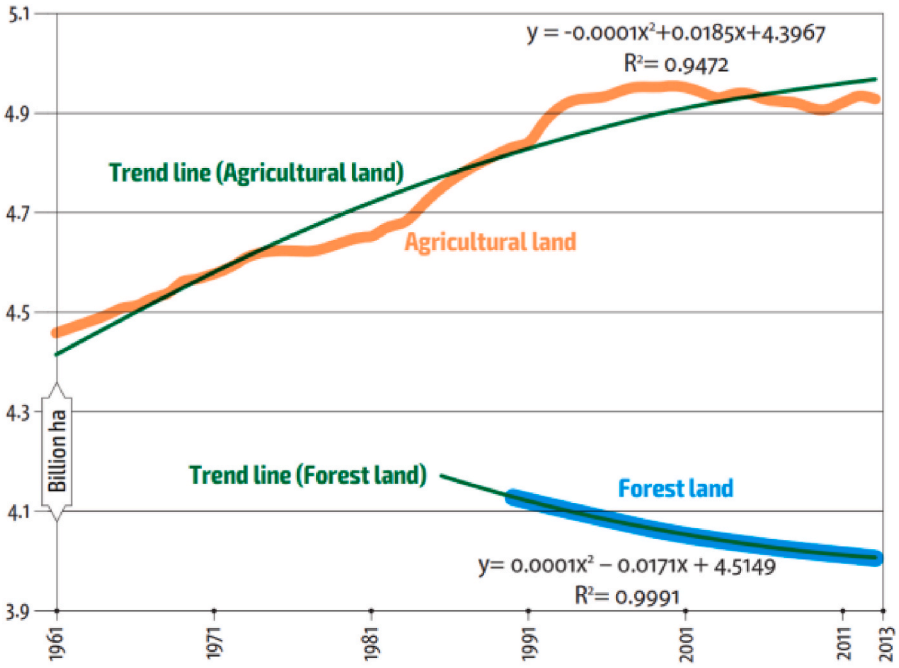


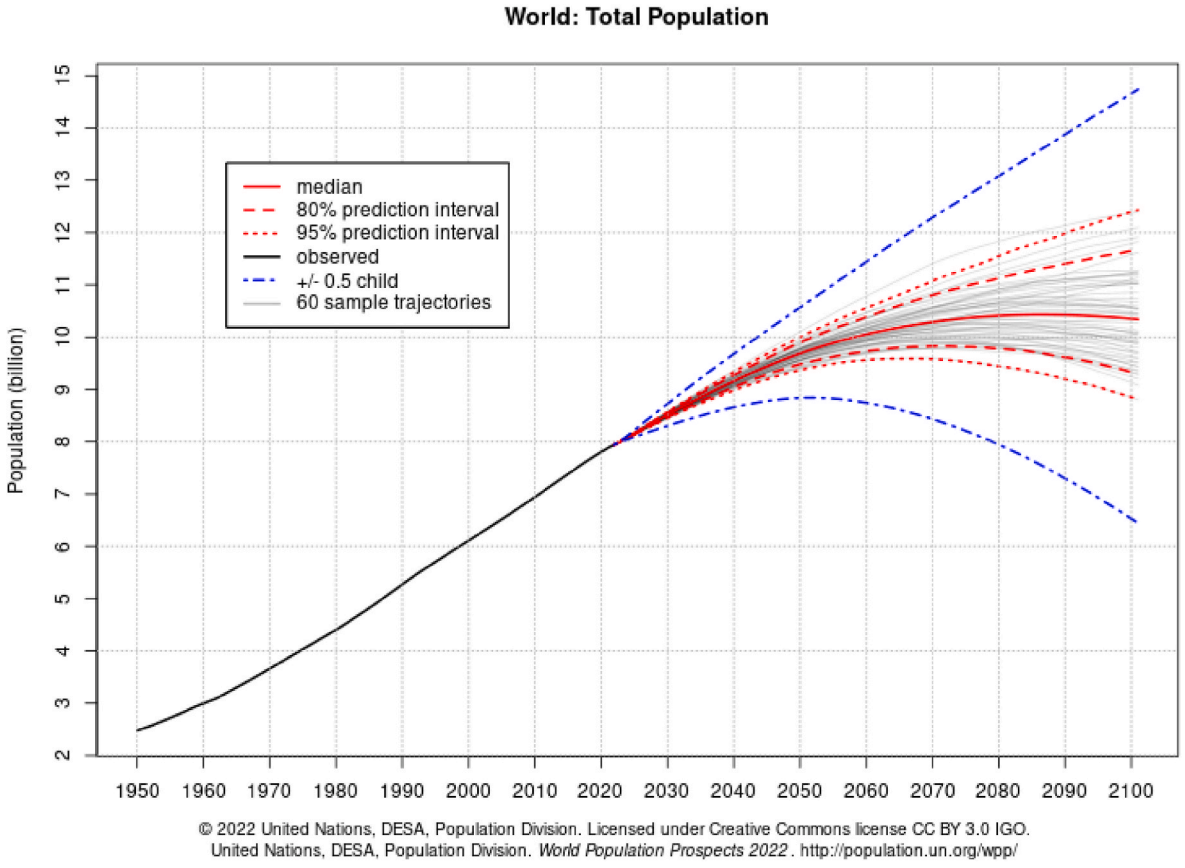
Fig. 1. Photovoltaic converter efficiency diagram according to National Renewable Energy Laboratory [24] (a) and world production capacity and cost of polycrystalline photovoltaic modules [25] (b).

The constant development of technologies and the increase in the standard of living of the Earth's population also contribute to the increase in the need for energy production, as well as the growth of demand for energy (Fig. 3a) [29–31]. At the same time, with the increase in the population, the growth of ag/cul production necessary for its nutrition also increases (Fig. 3b) [32].

The ever-growing demand for energy, food and water poses a significant challenge to achieving sustainable development, while the environmental situation is deteriorating year after year. Thus, today it is necessary to introduce more energy-efficient, resource-saving and «green » technologies, as well as increase ag/cul productivity while reducing the negative impact on the environment.

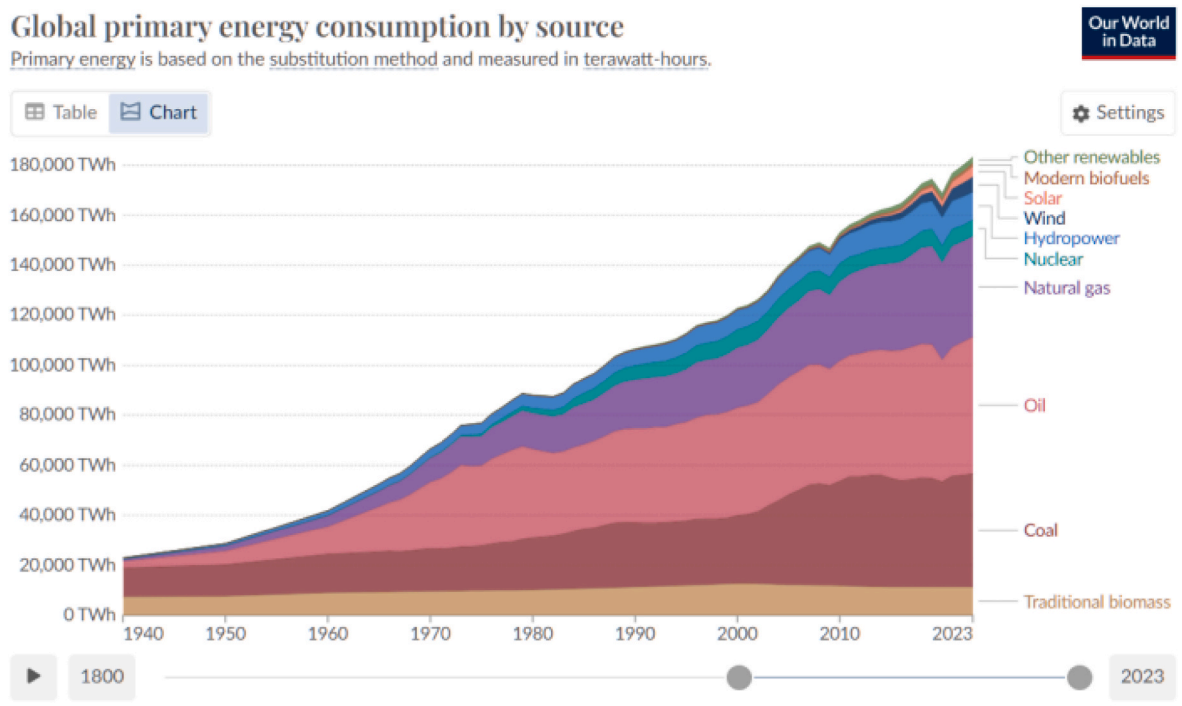


(a)

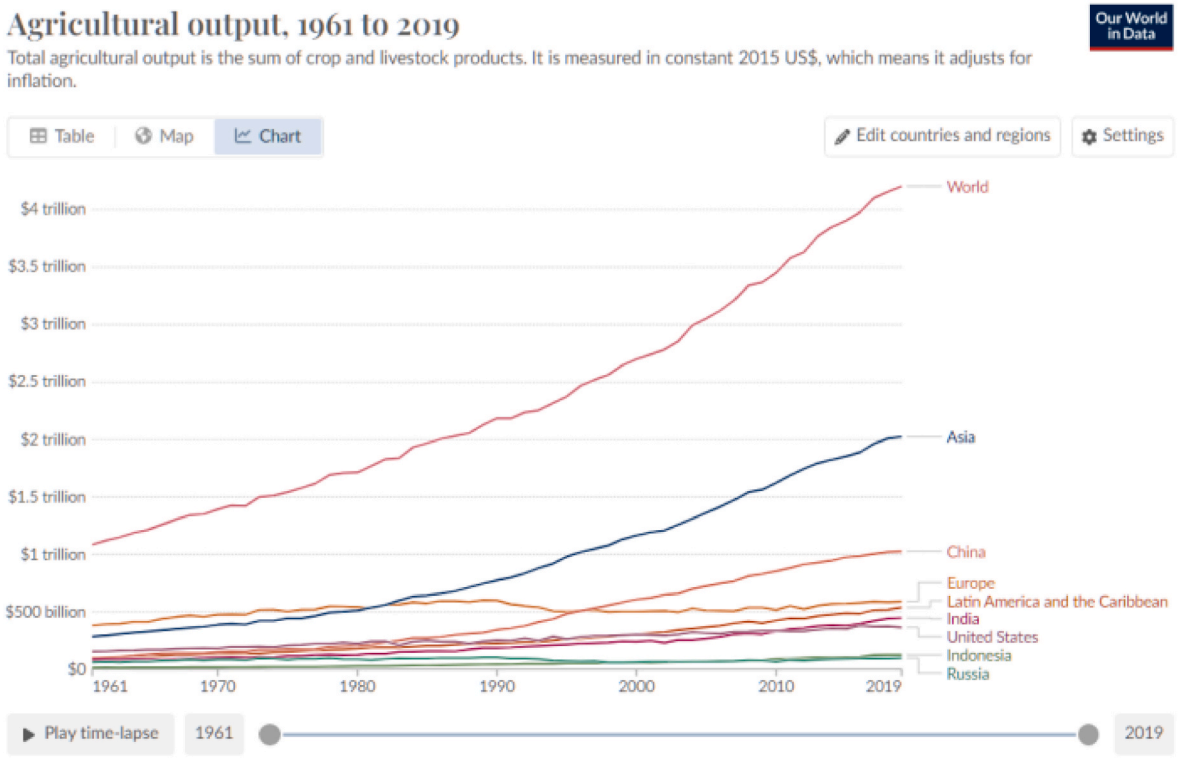


(b)

Fig. 2. Increase in agricultural land area and decrease in forest area [27] (a) and growth of world population according to UN data for 2022 [28] (b).



(a)



(b)

Fig. 3. Energy production growth from 1940 to 2013 [30] (a) and agricultural production from 1961 to 2019 [32] (b).

NOMENCLATURE

List of abbreviations and symbols	
Abbreviations and symbols	
AVS	Agrivoltaic system
AD	Anaerobic digestion

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List of abbreviations and symbols	
Abbreviations and symbols	
BGP	Biogas plant
RES	Renewable energy sourses
ICE	Internal combustion engine

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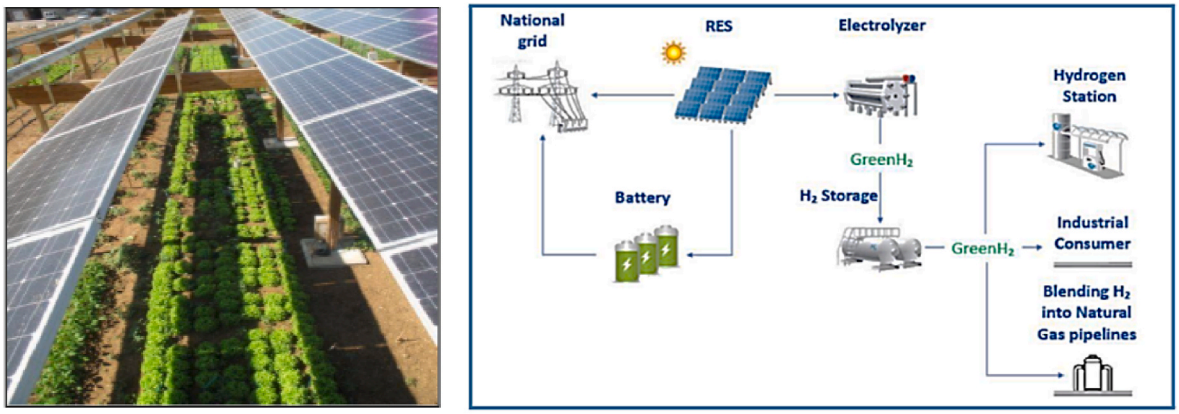


Fig. 4. Photovoltaic station above agricultural crops for the production of green hydrogen (a) and a general scheme for the production of green hydrogen in a similar system (b) [64]. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

(*continued*)

List of abbreviations and symbols	
Abbreviations and symbols	
GH	Green hydrogen
GHG	Greenhouse gases
OW	Organic waste
SR	Solar radiation
SC	Solar concentrator
SM	Solar module
SP	Solar plant
Ag/cul	Agricultural
USA	United States of America
SE	Solar energy
Te	Thermal energy
FC	Fuel cell
PVM	Photovoltaic module
PVC	Photovoltaic converter
PVP	Photovoltaic plant
EE	Electrical energy
UNO	United nations organization
y	Year
ha	Hectare
GW	Gigawatt
GW·h	Gigawatt hour
GW·h/ha	Gigawatt hour per hectare
g/m^2	Gram per square meter
kW·h	Kilowatt hour
KW·h/kg	Kilowatt hour per kilogram
km/h	Kilometer per hour
m	Meter
MW	Megawatt
MW/ha	Megawatt per hectare
Mil	Million
nm	Nanometer
t	Ton
h	Hour
°C	Degree Celsius
GPS	Global positioning system

2. Green hydrogen and photovoltaic technologies

One of the most promising types of fuel for energy, industry, transport [33,34], agriculture, and for the entire economy in general is hydrogen [35,36]. Hydrogen accumulation can become a worthy competitive option for the accumulation of EE using chemical power sources, when fuel cells (FC) are used in the system, and when using renewable energy converters, GHG emissions are minimized, and the cost of EE obtained in power systems in remote regions can be competitive compared to fossil fuel-based electric generators [37–40]. TE in such systems is expediently obtained by burning the resulting hydrogen, which will solve the issue of converting EE into TE when using chemical power sources in such systems.

Along with the direct combustion of hydrogen in internal combustion engines (ICE) [41–44], which produces a minimum of harmful emissions into the atmosphere, FCs [45–49] have high potential for implementation in transport, and they can become competitors to chemical batteries due to their disadvantages in transport [50,51] - high weight and cost, short range, disposal problems, etc. FCs have significantly fewer disadvantages (high cost, use of precious metals), but only long-term research and operation of hydrogen transport will help to determine niches for these two technologies.

It is difficult to compete in terms of environmental friendliness of production with GH [12,13,52,53], which is obtained by electrolysis of water (at the same time, valuable oxygen is also formed) [7–11], the EE for which comes from RES and, in particular, from solar photovoltaic power plants (PVP) [14,15], the efficient production of which is the focus of a large number of studies. Thus, research is being conducted to optimize the efficiency of a photovoltaic system for water electrolysis to obtain GH by comparing the voltage and maximum output power of a photovoltaic system with the operating voltage of proton exchange membrane electrolyzers [54]. Also, 19 hydrogen production technologies have been evaluated, their environmental impact, costs, energy and exergy efficiency have been considered, based on which the electrolysis technology operating with a photovoltaic system provides the highest energy and exergy efficiency among all photonic hydrogen production methods [55]. Both experimental and simulation studies have been conducted to evaluate the production of GH using the electrolysis process supported by a 20 kW PVP in Yazd, Iran: the developed system is capable of producing 373 t of GH per year [56]. Parametric studies of GH production considering climatic conditions in the Algerian Sahara regions were conducted to optimize the proposed solar system to maximize GH production [57]. GH production using SE in four different cities in Iraq was modeled and analyzed, and the best potential location for GH production was proposed based on solar radiation (SR) characteristics and ambient temperature [58]. The GH system is also proposed to be produced in Indonesia using 3718 lithium-ion batteries, each with a nominal capacity of 210 kW h, and a proton exchange membrane electrolyser powered by a floating PVP [59]. A power management strategy has been developed to organize the energy flow for hydrogen production in a grid-integrated photovoltaic system [60], in which hydrogen production is optimized by managing the flow of water.

One of the alternative options for possible conservation of water and land space is floating PVP, where saving land space is beneficial for the environment and attractive from a business point of view [61]. At the same time, there is a growing positive outlook for integrated photovoltaic FC systems as new clean energy technologies. Hydrogen

generation from excess EE provides a long-term strategy for energy storage, which can also be applied in industrial processes and chemical synthesis, in addition to its use for EE generation [62].

Agriculture, PVM and GH can be combined to create a powerful synergy for sustainable human development when such an agro-engineering complex can generate EE to power the electrolysis process for GH production, creating a closed-loop clean energy system. By using the same land space for both PVP and agriculture, such a system can free up valuable space for additional GH production facilities. Due to the flat and sealed design of the PM, such a system allows for centralized collection of rainwater [63], which can then be used for irrigation of ag/cul crops, electrolysis to produce GH, etc. Thus, PMs are also a protective structure for agricultural crops from precipitation that can damage plants. Along with hydrogen transport, it is advisable to sell and use GH as an accumulation of EE when using FC. The integration of agriculture and GH can provide flexibility and resilience to the grid, while excess EE from SR can be used to produce GH, mitigating grid overload during peak generation periods, and the stored GH can be used to generate EE when SE is unavailable. At the same time, GH can also be used to generate TE for the ag/cul complex by directly burning it in boilers or ICEs with simultaneous generation of EE and TE.

Current research is investigating possible measures to reduce the costs of water electrolysis using a ground-based PVP with a single-axis tracking system for the position of the Sun over ag/cul crops for the production of GH, which considers the annual productivity of such a system, the production of GH and its consumption by the electrolyzer [64] (Fig. 4).

Based on current research, there is a need for further optimization of PVP designs in agriculture for GH production: optimal system configurations to maximize both yields and EE generation while minimizing land use conflicts [65]. Cost reduction is also needed: the cost of GH production needs to be further reduced to achieve widespread adoption, while technological advances in electrolysis and infrastructure development are critical [66]. Supportive political will and legislation are also needed to stimulate the development and implementation of PVP technologies in agriculture for GH production [67].

At the same time, the use of PVP in agriculture for the production of GH represents a compelling approach to addressing the interrelated challenges of energy, food and water security. By harnessing the potential of SE and promoting a symbiotic relationship between agriculture and clean energy production, we can move towards a more sustainable future for humanity. Modern research and innovative approaches can address the underlying challenges of the technologies discussed above and are key to unlocking the full potential of the symbiosis of agriculture and photovoltaics for the production of GH.

3. Introduction of photovoltaic converters in agriculture

The location of PVM for the production of «green » EE is advisable both on the roofs of buildings in the form of standard PVM or roofing and siding panels for the simultaneous protection of buildings, and on the ground in the form of large PVM fields, however, the location of such PVP requires the allocation of a large amount of space on the ground, which is eliminated from the construction and ag/cul industries. At the same time, the constant population growth and the increase in land area for ag/cul production pose a choice between the feasibility of using land for energy generation or food production, when agro-food processes use almost 30 % of the world's total energy, accounting for about 19–29 % of annual global GH emissions, while ag/cul activities alone contribute to about 20 % of annual total carbon dioxide emissions [68,69], when energy production in ag/cul processes makes the largest contribution to carbon dioxide emissions, equal to 78.7 % [70]. Optimization of farming processes, automation, mechanization and more efficient use of equipment and resources can make the ag/cul sector more sustainable as it can increase production and income, which will increase the energy demand for ag/cul activities [25].

Using the space under the PM for growing various ag/cul crops or breeding livestock will improve the economics of the projects, provide autonomous energy supply in remote regions, produce environmentally friendly products, improve the growth of plants and animals by regulating shading, evaporation and other processes. Solar rooftop PVP technologies can be a solution to the problem of competition for land between ground-based PVP and specialized bioenergy crops [71]. At the same time, there is an increase in economic value by more than 30 % when EE production is associated with the cultivation of shade-tolerant crops [72]. In such a layout, PM slightly interferes with the growth and yield of ag/cul crops due to the increase in shade, but the shading area can vary depending on the location of the PM and the crops themselves. Although such a symbiosis of SE and agriculture was initially proposed back in the 1980s - Agrivoltaics [73], since then SE technologies have improved significantly, while the cost of installed capacity has decreased significantly.

Traditional approaches in energy and agriculture often prioritize one technology over another, leading to conflicts over land use and environmental degradation. The combination of agrivoltaics and GH technologies offers a promising solution through a synergistic approach that uses SE for both EE and clean fuel production, contributing to a more sustainable future. The main objective of such a symbiosis of technologies is to achieve net zero emissions by 2050 [74], which forces countries and high-level decision makers to prioritize the development of renewable energy as a key tactic in the fight against climate change [75]. To cope with the growing demand for food, ag/cul intensification is inevitable, which also increases energy demand when fossil fuels can be replaced by renewable energy, while PVP capacity is growing exponentially and EE costs are at a minimum in many countries [76,77]. It is projected that RES will surpass coal in energy generation by 2050, accounting for 50 % of global energy production. The potential of SE in agriculture and mandatory energy supply measures will further enhance its role [78,79].

Integrating PVP technologies into agriculture can significantly mitigate the damages of climate change by creating a new energy-food nexus that will benefit both agriculture and energy [73]. The deployment of agrivoltaic systems (AVS) worldwide with a capacity of over 15 GW has the potential to increase farmers' income and make it more sustainable in a variety of ways [76]. GH is also expected to play a significant role in achieving these goals [59,80]. Thus, AVS can be a valuable technical approach for more sustainable agriculture, helping to meet current and future energy and food production needs while conserving land resources [81], offering environmental benefits, commodity production, and income opportunities for low-margin agriculture. However, current methods do not fully exploit this symbiotic energy to improve ag/cul production, although improvements can include smart farming, growth stimulation, fertilizers, pesticides, fuel reduction, on-site production, and disaster recovery [76].

Providing ag/cul crops with the necessary radiation spectrum for their growth, along with minimizing the negative impact of shading on ag/cul crops, will improve their growth and increase the amount of output, which will also have a positive effect on the project's economy. However, it is necessary to take into account that shading of ag/cul crops in such ABCs can lead to deterioration in growth or even loss of crops, but also provide protection and increase the yield or quality of agricultural crops in certain climatic conditions.

During the operation of such an AVS, GHG emissions are reduced, and ag/cul and energy become « sustainable». However, when implementing such systems, it is necessary to solve problems of optimizing the design of the PVP: the size of the PM, their angle of inclination, the distance between rows, etc., as well as the varieties and conditions of growing ag/cul crops, types and conditions of keeping ag/cul animals [82]. The implementation of AVS has increased exponentially in recent years and reached a global installed capacity of 2.8 GW in 2020, compared with an initial capacity of 5 MW in 2012 [83].

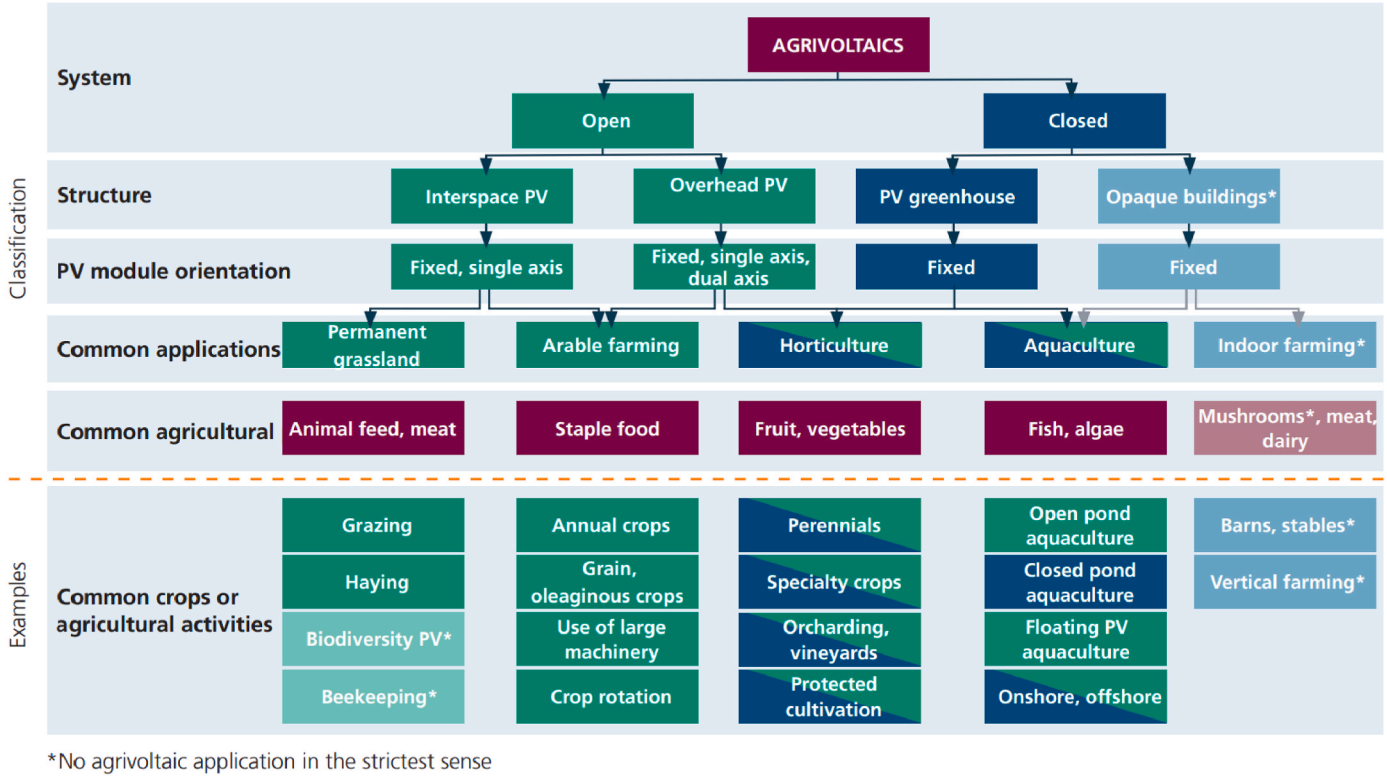


Fig. 5. Classification of agrivoltaic systems according to various system parameters [84].

4. Classification of agrivoltaic systems

AVS can be classified according to various parameters (Fig. 5) [84], such as system type, design type, module inclination, application type and ag/cul products grown.

Open AVSs use interval PMs with a fixed tilt angle or with a single-axis solar tracking system, which are installed over meadows, pastures or cultivated lands where typical ag/cul products such as feed and meat and dairy products, as well as other staple foods on cultivated lands, are grown. Also, the main food products on cultivated lands are grown in open AVSs with overhead PMs with a fixed PM tilt angle or a single- or dual-axis solar tracking system. Closed AVSs use greenhouses with PM and opaque buildings with a fixed PM tilt angle, where garden plants, fruits and vegetables are grown, as well as open AVSs with overhead PMs with a fixed PM tilt angle or a single- or dual-axis solar tracking system. Also, in closed AVS with a fixed angle of inclination of PM, as in open AVS, aquaculture in the form of seafood can be grown. In AVS with opaque buildings, ag/cul production is possible, intended for indoor (intra-house) cultivation of meat and dairy products and mushrooms.

In the considered AVSs, each system and its design have their own typical ag/cul activities and ag/cul crops. Thus, for an open AVS, cattle grazing, haymaking, improving and promoting biodiversity, beekeeping, growing annual crops, temporary pasture, crop rotation, using large-sized equipment, aquaculture in an open pond and aquaculture under floating PM are appropriate. For open and closed AVS, perennial crops, permanent crops, special crops, gardening and growing aquaculture under coastal or marine PM are relevant, while only for a closed AVS, growing aquaculture in greenhouses, ag/cul barns and stables, as well as vertical cultivation of ag/cul crops are suitable.

4.1. Open agrivoltaic systems

Based on global forecasts and trends in the development of photovoltaic energy [85], we can conclude that there is a growing trend towards increasing the use of double-sided PVPs compared to single-sided ones and increasing the use of PMs built into buildings and their wider

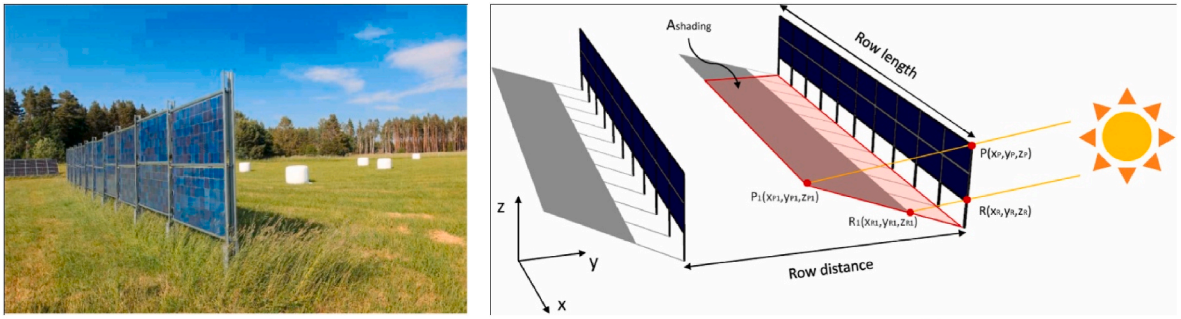


Fig. 6. Vertically mounted agrivoltaic system with bifacial PV modules installed in Karrbo Prästgård, Västerås, Sweden (a) and calculation of the shaded area for shading coefficient estimation (b) [90].

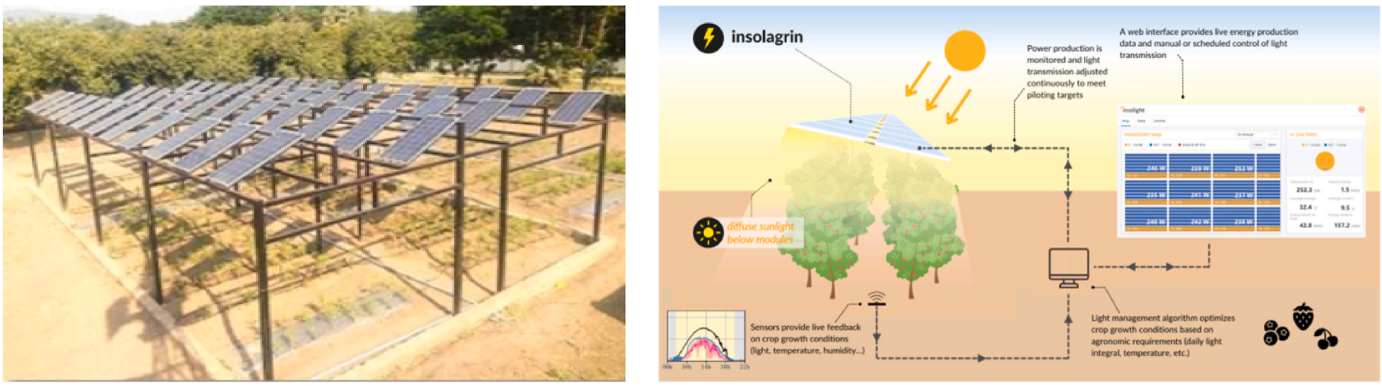


Fig. 7. Agrivoltaic system with staggered configuration of photovoltaic modules in Gujarat, India [96] (a) and Insolagrin - an agronomic tool for crop protection and simultaneous production of electrical energy [97] (b).

use in agriculture. Vertically mounted PM arrays are mainly oriented to receive SR from east to west, are installed close to the ground surface and have maximum power in the morning and evening [86,87], which also solves the problem of mismatch between the demand for EE and the EE generation profile of PM arrays oriented north/south [88], while open mounted PM arrays can have any orientation to receive SR. When using fixed tilt and north/south facing PMs, there is spatial heterogeneity in the daily distribution of sunlight to crops and high soil moisture content, which may affect crop yields [89].

Two-sided vertical PMs provide higher crop yields, while one-sided PMs with a fixed angle of inclination to the horizon and directed north/south provide higher EE production [89].

The distance between rows of vertical bilateral PMs significantly affects the distribution of photosynthetically active radiation, and at the latitude of Sweden, the yield of potatoes and oats can be reduced by almost half by changing the distance between rows from 20 to 5 m, while the optimal distance between rows is 9.7 m for potatoes and 9.2 m for oats (Fig. 6) [90].

The vertical arrangement of PMs oriented to receive SR from east to west is appropriate for perennial ag/cul crops, while from south to north is appropriate for growing shade-tolerant ag/cul crops in the summer, when EE production is a priority. With the introduction of PMs in agriculture, land productivity increases by 50 %, but EE production decreases by 33 % compared to traditional and separate production of EE and ag/cul products [91]. Vertical two-sided PMs oriented to receive SR from east to west provide a much better distribution of illumination during the daytime compared to PMs with a fixed angle of inclination to the horizon and directed to the south, which indicates greater suitability for growing ag/cul monocultures [92]. Such vertical two-sided PMs provide the lowest ground coverage, reduced construction costs, low obstruction to ag/cul machinery and rainwater, lower visual impact on the landscape, mitigation of wind impact at crop heights, and reduced PM pollution [92,93].

Overhead PM, in turn, provide greater protection of ag/cul crops from unfavorable SR, precipitation and temperatures, however, to produce the same amount of EE, compared to standard PVP, 20–40 % more land area will be required (for example, for the USA this is 0.6 MW/ha and 0.87 MW/ha, respectively) [84,94,95]. Whereas in Gujarat, India, an AVS elevated at a height of 3 m with a PM electrical efficiency of 11.22 % generated 12667.15 kW h from 2017 to 2018 and was used to study shade-tolerant tomato crop where its yield increased by 15.09 % in such AVS [96] (Fig. 7a). Solar tracking PMs in AVSs can mitigate seasonal crop yield variability associated with climate change and can improve spatial uniformity of crop production while reducing crop water requirements, but such systems can increase the cost and complexity of AVSs [85].

AVSs with translucent PMs provide control of light transmission, where the optical layer is continuously controlled to deliver a precise amount of light to crops; protect crops from rain, snow or hail, where rainwater can be collected through gutters for irrigation; provide ventilation, which is provided by openings in the structure to maintain optimal temperature and humidity conditions; a night mode of operation protects against frost, where the optical layer automatically opens at night to keep the crop warm; reduce water consumption with the help of an optical layer that shades crops, keeping them cool during extreme heat (Fig. 7b) [97].

Research is also being conducted to explore the feasibility of using open AVS in existing grape farms in India, using shade-tolerant crops as an example, where installing PM between trellises in grape farms can increase the economic value by more than 15 times compared to conventional farming while maintaining grape production. The implementation of such dual land use can generate more than 16000 GW h of EE, meeting the energy needs of more than 15 million people [98].

Along with crystalline PVMs, thin-film translucent modules are also used in AVSs. They have a low mass per unit area (about 500 g/m^2) [84], an aesthetic appearance, uniform transparency and a better temperature coefficient compared to crystalline silicon PVMs, but are rarely used in open AVSs due to their relatively higher cost and lower productivity [83]. In open AVSs with overhead PM modules, this type of PVM is installed at a height of 4–7 m above the ground [99].

Some plants, such as lettuce, have the ability to adapt to shade conditions and partially or completely compensate for reduced light [100] through morphological changes in leaf development and morphology. An AVS with a low-density PVM design installed over corn, a typical shade-tolerant crop, increased yield by 5.6 % compared to a control configuration without PVM [101]. This result may be due to several effects of the interaction between the environment, plants and PVM: only a small fraction of the falling SR is required by plants to achieve the maximum rate of photosynthesis, while eventually a level is reached at which SR is no longer a limiting factor for the overall rate of photosynthesis, when too much SR hinders crop growth, and daily exposure to strong ultraviolet radiation can cause serious damage to plants when plant defense mechanisms from sunlight in the outer layer of leaves are triggered; at the same time, shading reduces water evaporation, reduces diurnal variations in crop and soil temperatures, and reduces soil erosion [101,102]. Cooling of PVMs due to the cultivation of ag/cul crops under them and the retention of moisture also allows to increase their efficiency by 10 % [103,104]. It has also been established that the daily air temperature does not change under PVMs, which also reduce the daytime and nighttime temperature amplitude of ag/cul crops and soil, while the growth rate does not decrease under PVMs [105,106].

The implementation of AVS may require minor adaptations in farming practices. It should also be noted that the intrinsic efficiency of

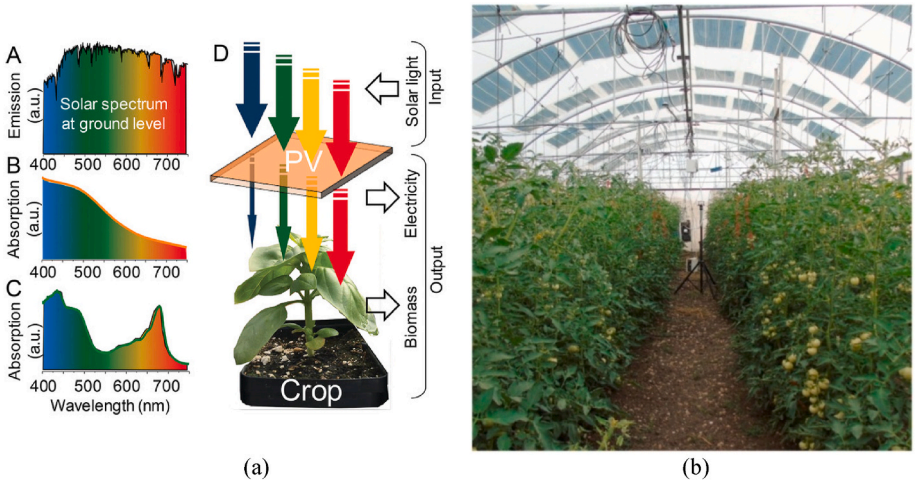


Fig. 8. Solar irradiance spectrum in the visible range at ground level (A), absorption spectrum of a tinted translucent photovoltaic module (a-Si) (B), absorption spectrum of a basil leaf (D) [110] (a) and a greenhouse with installed organic photovoltaic modules [112] (b).

the photosynthetic process is quite low, only about 3 %, while commercially available monocrystalline PVMs have electrical efficiencies of about 20 %, while for different PVM installation densities, global land productivity is predicted to increase by 35–73 % [107]. AVSs can be optimized both by crop selection and PVM placement to find the best compromise between food production and EE production on the same plot of land.

Land use efficiency in AVS can reach 160 % and 186 % compared to separate PVP and cultivation of crops alone [84]. Generation of EE from SR in combination with cultivation of shade-tolerant crops leads to an increase in economic benefit by more than 30 % for farms (using USA residential EE tariffs as the cost of EE generated by SE) using AVS instead of traditional cultivation of shade-tolerant crops [108]. As a result, the capacity of the PVP system will increase from more than 40 GW–70 GW when lettuce cultivation in the USA will be accompanied by EE production in the AVS [108]. The effect of changing the tilt angle of the PVM during the day has also been studied, where high productivity per unit land area can be achieved using a sun tracking system rather than stationary PVM systems, where lettuce yields were found to be close to those under full sunlight conditions due to physiological adaptation of plants to temporary shading conditions, resulting in higher SR utilization efficiency of the plant [109].

4.2. Greenhouse agrivoltaic systems

Most of the PVMs consist of crystalline silicon PVCs, however, other types of PVCs are also used in greenhouse AVSs and have their own advantages and disadvantages. Thin-film translucent PVMs, such as CIGS, CdTe, a-Si and micro a-Si, have a low mass per unit area (about 500 g/m^2) [84], an aesthetic appearance, uniform transparency and a better temperature coefficient compared to crystalline silicon PVCs (Fig. 8a) [110]. Thus, amorphous silicon-based translucent photovoltaic glass slightly reduces the production of vegetative mass and accelerates the apical growth mechanism of heliophilic plants, but from a statistical point of view this effect is insignificant, so such a technology is viable for horticultural production when the energy balance carried out shows that the EE generated by the PVM exceeds the EE consumed by the greenhouse [111].

Organic PVCs are also a promising direction of agrivoltaics due to their low cost of production, however, the limiting factors for their widespread use today are the effects of oxygen and water, irradiation and heating, metastable morphology, diffusion of electrode materials and buffer layers, as well as mechanical stress [99]. At the same time, the installation of organic PVCs inside a greenhouse can reduce the rate of their degradation, and the absence of fastening the modules directly to the polyethylene covering of the greenhouse allows avoiding loads on the module due to film movement, which will also reduce the degradation of the modules (Fig. 8b) [112].

The results of modeling a greenhouse with organic translucent PVCs with an EE conversion efficiency of 9.4 % and an average transmittance in the visible range of 24.6 % show an increase in the dry matter mass of tomato crops with translucent organic PVCs by 46 % compared to conventional agrivoltaic greenhouses with silicon PVCs [113].

Dye-sensitized PVC is an electrochemical device that uses light-

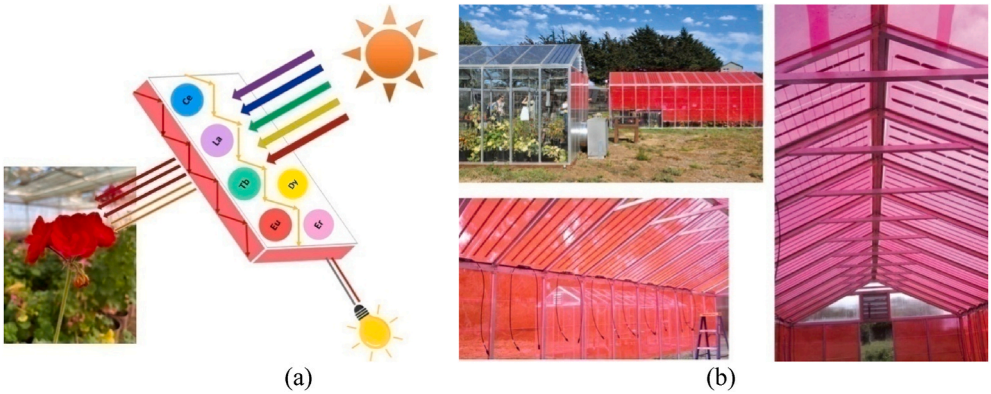


Fig. 9. Operating principle of a luminescent solar concentrator in agrivoltaics [115] (a) and a greenhouse built using wavelength-selective photovoltaic modules [116] (b).



Fig. 10. Semi-flexible photovoltaic modules based on polymer film (a) and translucent photovoltaic modules based on double-sided glazing with monocrystalline photovoltaic converters for greenhouse agrivoltaic systems (b) [118].

absorbing dye molecules adsorbed on semiconductor nanoparticles to generate EE from SR when favorable lighting conditions are provided in the greenhouse by using novel sensitizers with improved transmittance of red (625–675 nm) and blue (425–475 nm) wavelengths [114]. Greenhouses with such SR converters can improve the thermal stability of the greenhouse by partially blocking infrared radiation and increasing biomass yield compared to conventional greenhouse glazing and opaque photovoltaic greenhouses, their performance is independent of the angle of incidence of light, but their stability and efficiency are limited [83].

Perovskite PVCs can also provide translucency or absorption of different wavelengths, which makes their combination with crystalline silicon PVCs more efficient for using the solar spectrum, while the electrical efficiency of perovskite PVCs is steadily increasing (more than 25 %) [24], and the fight against rapid degradation and scalability is leading to certain successes.

Luminescent solar concentrators (SCs) based on rare earth complexes allow photosynthetically active radiation to reach plants, promoting efficient growth of ag/cul crops during photosynthesis, while non-photosynthetically active radiation (usually light outside the 400–700 nm range) can be directly captured by them and converted into EE without affecting the photosynthesis process [115] (Fig. 9a). Compared with organic dye-doped fluorescent SCs, fluorescent SCs based on rare earth complexes mixed with various host polymers are transparent, which is important in greenhouses to provide daylight and low energy consumption. Wavelength-selective photovoltaic systems combine fluorescent PVC technology with conventional silicon-based PVCs, thereby increasing the efficiency and reducing the cost of energy generation when a portion of the blue and green wavelengths of the solar spectrum is absorbed, and the remaining wavelengths are used for photosynthesis by plants. The effect of tomato photosynthesis in similar greenhouses showed a small reduction in water consumption [115], and a greenhouse constructed using wavelength-selective PVMs consisting of polymethyl methacrylate plastic impregnated with the fluorescent dye perylene red, produced EE via narrow photovoltaic strips (black lines on red panels) sufficient for greenhouse fans, microclimate monitoring sensors, and data loggers [116] (Fig. 9b).

The absorbed energy is transferred inside the colored plastic and goes to the narrow PVCs, which provides an increase in power by two or more times per unit area due to the placement of wavelength-selective fluorescent absorbers [116].

Another new type of PM for greenhouse AVSs are organic PVCs, which are capable of mitigating the undesirable shading effects of traditional opaque PVCs and regulating the light spectrum [117].

Although the above-mentioned PVC technologies are also used in greenhouse AVSs, the most widespread today are PVMs with crystalline PVCs, which can be semi-flexible and repeat the profile of the roof or have double-sided glazing in their design and an incompletely filled PVC surface [118] (Fig. 10).

Such PVMs, occupying 20 % of the greenhouse roof area, reduce SR by 35–40 % compared to the use of a polyethylene cover, which is within the permissible range of photosynthetically active radiation for lettuce plants, while such shading reduces air temperature by 1–3 °C and does not affect relative humidity during natural ventilation, and shading slightly increases wet weight, leaf area, and chlorophyll content [119]. Thus, such greenhouses will compensate for part of the EE consumed by the greenhouse and provide better lighting conditions for shade-loving plants such as lettuce, spinach, carrots, turnips, cabbage, and broccoli during the hot season.

4.3. Closed agrivoltaic systems

Closed AVSs are not the main direction of AVS development, although they allow growing a wide range of ag/cul plants and animals, as well as providing various ag/cul processes, such as the production of GH from ag/cul waste. In such systems, cultivation and ag/cul processes take place in closed spaces, on the roofs and walls of which solar modules (SM) of various designs are installed [120].

PVC of SR are used to supply electricity to small devices in various systems and to monitor various processes in a biogas plant (BGP). The cost of a large installed capacity of such PVM is still high today, and the

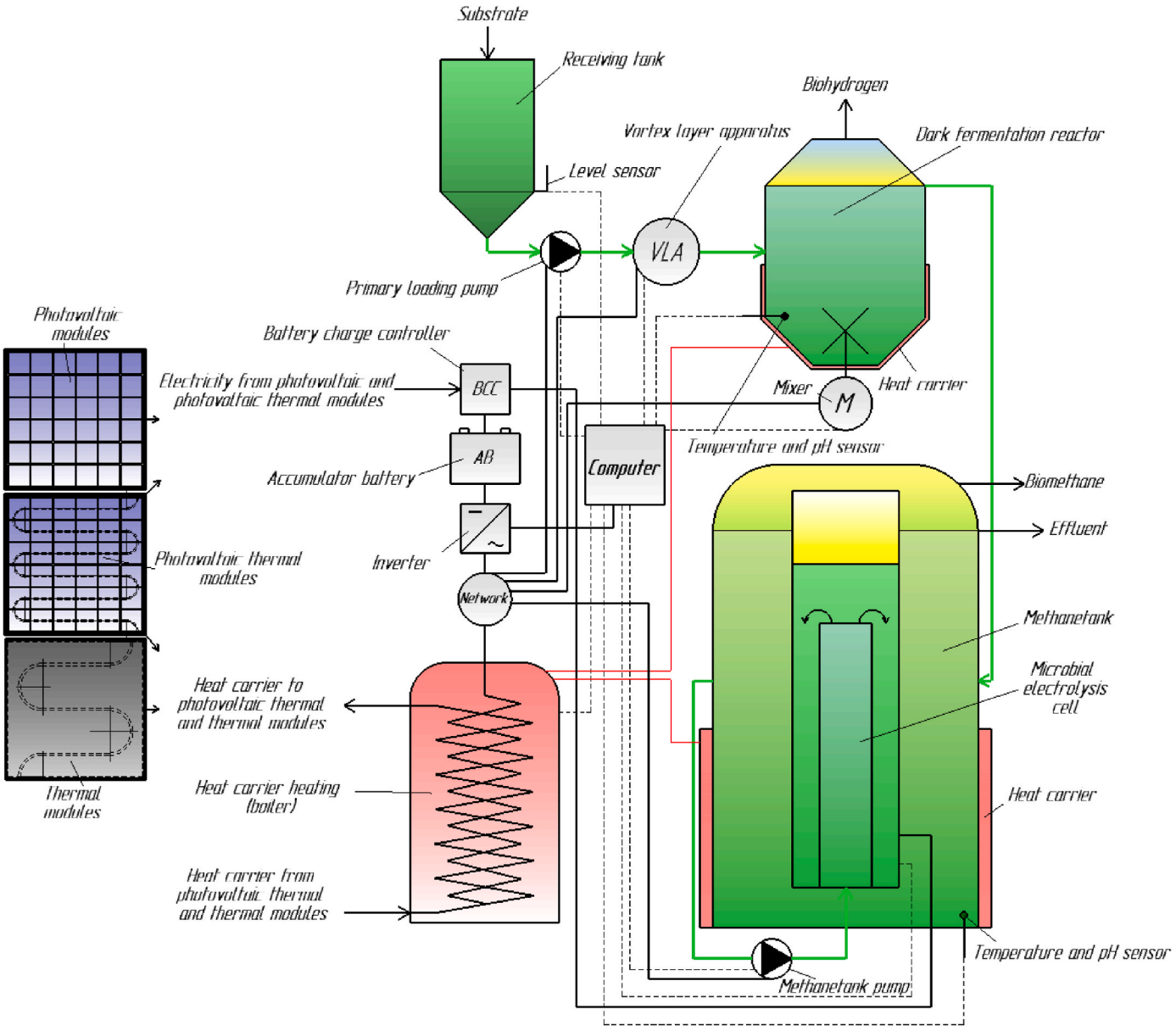


Fig. 11. Technological scheme of a biogas plant with parallel energy supply from solar modules of various types [133].

electrical efficiency is low (about 15–20 % in serial modules) for supplying powerful equipment, however, in remote regions without a centralized electricity supply and for supplying power to low-power systems, such devices are used [121–125]. Today, the cost of producing GH is still higher than other types of hydrogen obtained without the help of RES, but the cost of its production technologies, as well as the cost of renewable energy converters and, in particular, PVM, is steadily decreasing [126–128]. Providing such anaerobic bioconversion systems with energy from SR converters for the purpose of obtaining biohydrogen and biomethane will increase the economic efficiency of such systems by reducing the use of produced gas for energy supply for their own needs, which will make such systems cheaper and more efficient to operate. The use of GH in BGP, also obtained from EE generated using PVM, also improves the quality of biogas [52,53], when GH binds carbon dioxide and brings the quality of biogas closer to that of natural gas (biomethane) [129–132].

In order to supply energy to the BGP in parallel operation with the existing network, a scheme for supplying heat and electricity to the BGP using SM of various types (photovoltaic, thermal, photovoltaic thermal) [133] (Fig. 11) has been proposed, which are located on the surfaces of a container, which is also a closed AVS (Fig. 12).

In the proposed closed AVS, using different types of SMs, it is possible to obtain the necessary heat and electricity supply for the BGP by combining the composition of the SMs, when the designs of the proposed SMs [134] ensure the versatility of their location - on the wall of a building, as a siding panel, on the roof of a building, as a standard SM, as a separate SM field in the form of a ground-based SP, as a result of the operation of which energy, biohydrogen, biomethane and high-quality fertilizers are produced. When locating such a containerized closed AVS, in addition to the recommended orientation to the cardinal points (south), it is also advisable to consider the orientation of the receiving surface to the east and southeast in order to shift the maximum energy generation from the SR to the morning hours to maximize the convergence of energy generation and consumption schedules.

Such agrivoltaic containers can also accommodate vertical farms, poultry houses, fish ponds, mushroom and insect growing racks, etc. The generated EE can be used both directly to power the agrivoltaic container and to generate GH using water electrolysis, which will be an

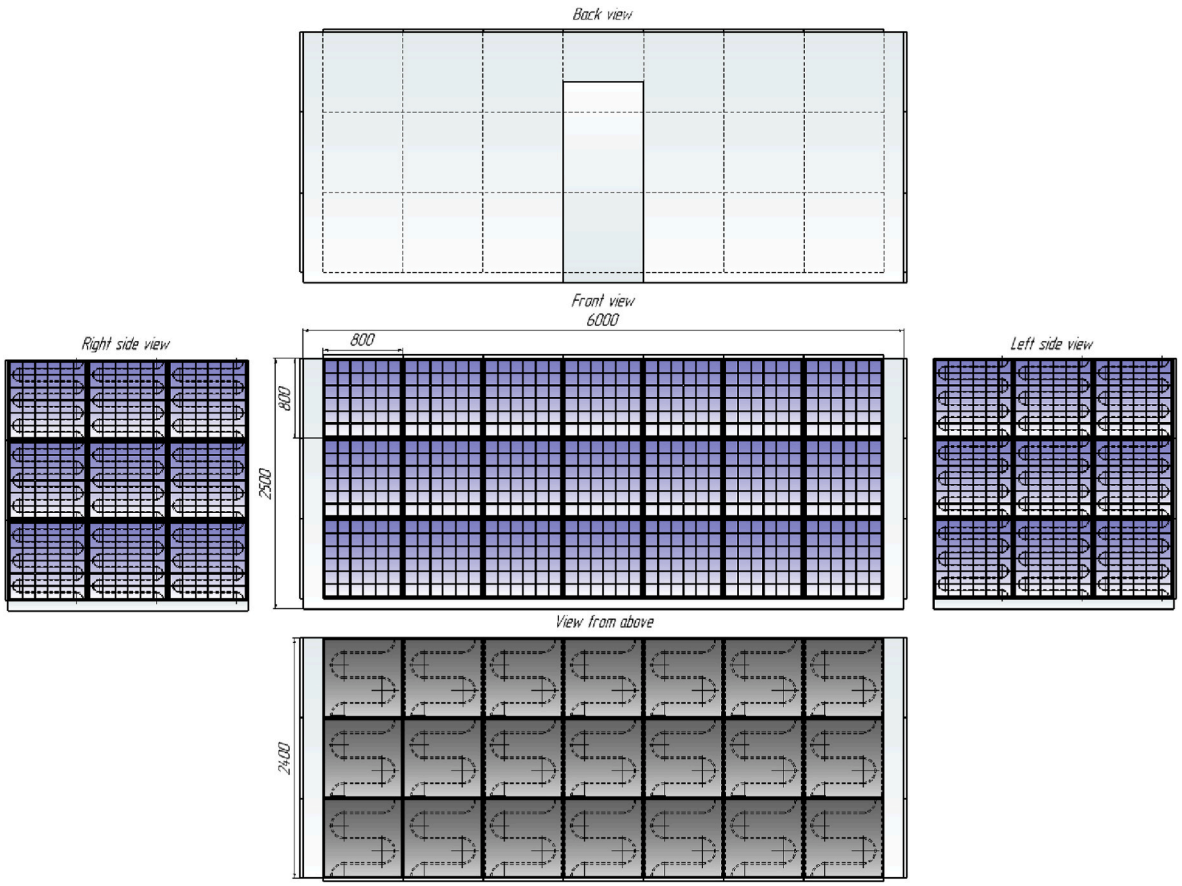


Fig. 12. Full coverage of standard container surfaces with solar modules [133].



Fig. 13. Electric tractor with photovoltaic modules [25] (a–d) and small-sized agricultural machine with photovoltaic modules [137] (e).

energy storage device and can be used for various purposes. It is also advisable to obtain water for electrolysis using EE from PVM from atmospheric air.

5. Agricultural machines and autonomous robots in agrivoltaic systems

The considered AVSs of various types imply the use of various auxiliary technical equipment in the form of ag/cul machines and autonomous robots, the development and research of which is actively carried out throughout the world. Electricity supply of such devices can occur with the help of PVM, EE from which can be accumulated both in chemical batteries of EE, and with the help of GH in reservoirs.

One of the difficulties in the implementation of AVS is the accumulation of EE, for the storage of which both batteries and flywheels, gravity accumulation and supercapacitors can be used. However, the most promising tool for storing EE and TE in AVSs may be GH, which can be used both in ag/cul transport and in FCs for the generation of EE, as well as in electric generators for the generation of TE. With an average electrical efficiency of 17 % with a capacity of PVM of AVS of 0.6 MW/



Fig. 14. EcoRobotix robot [138] (a), autonomous Ladybird spraying robot in Cowra, New South Wales [139] (b), RIPPA robot on a lettuce farm in Lindenough, Victoria [139] (c).

ha, their generation of EE is 0.09 GW h/ha, when, taking into account the average production of GH of 55 kW h/kg for most electrolyzers, it is possible to obtain 16 tons of GH from 1 ha per year [82].

In ag/cul electric tractors, PVMs are installed in the on-board charging systems [25], which supply power to electrical devices for various purposes [135,136]. A prototype of a light-duty hybrid electric tractor with PVM for use on farms provides almost 18 % of the daily EE demand using PVMs, with the remaining EE provided by the local power grid [25] (Fig. 13a–d). The single-row moldboard plough and the two-row seeder can work continuously for about 4 h a day thanks to the 16.5 kW h battery. The standard mower, sprayer and fertiliser module can work for 3.2, 3.6 and 3.4 h a day, respectively. The tractor can also move a two-ton trailer for 2 h on asphalt at a speed of 25 km/h and move for 2.5 h at a speed of 18 km/h on a sandy road, as well as for 3 h at a speed of 9 km/h on an ag/cul field [25]. It is also advisable to mount the PVM in various small-sized ag/cul equipment for various purposes (Fig. 13e) [137].

The ecoRobotix autonomous sprayer robot with PVM performs crop weeding operations [138] (Fig. 14a), uses a GPS receiver and camera to position itself on the farm and track crop rows to detect weeds between them for micro-dose herbicide application. The robot can treat a maximum of 3 ha per day and 10 ha per week at wind speeds below 60 km/h, and commands can be transmitted from a smartphone [25]. The Ladybird robot can operate for 7–9 h per day, and has various sensors for hyperspectral, thermal, infrared, panoramic vision, stereo vision with strobe, lidar and GPS [139] (Fig. 14b). In the autonomous vegetable weeding robot RIPPA, the platform arrangement is changed to make the structure lighter compared to Ladybird, while it can provide reliable and more convenient operation on the farm [139] (Fig. 14c).

Despite numerous studies and implementations of ag/cul machinery with PVM in AVS, the main problems preventing wider implementation of such ag/cul machinery are high initial costs, mainly associated with batteries, and shortcomings in EE storage technologies [25,140]. Thus, the use of GH for EE storage in such devices is a fairly promising direction and can solve the issues of EE storage.

6. Conclusion

Thus, the use of PVMs produced using various technologies finds wide application in agriculture. The EE they generate can be used both in greenhouses, where PVMs are built into the greenhouse structure, and in fields and open spaces, where ag/cul machinery with installed PVMs and recharged using stationary PVMs can be used. At the same time, the use of GH for the accumulation of EE in ag/cul transport is quite a promising direction. The use of PVMs in BGPs allows reducing the costs of EE and producing GH from OW. It is advisable to use a container layout of such closed AVSs to simplify the installation of PVMs. Increasing the production of GH is currently a priority for most developed countries, since such a task is consistent with the vision of their economic, energy and political independence, and is also an important component of the fight against climate change. The use of EE generated by AVS in water electrolysis to produce GH opens up great opportunities for countries that want to diversify their economies with a large share of the ag/cul sector.

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